EXECUTIVE SUMMARY • NON-TECHNICAL BUSINESS GUIDE

Under 40 Report Dashboard

Contractor Weekly Hours Merging, Leave Audit & Compliance Tool

AUDIENCE	CORE GOAL	VALIDATIONS	DATA PRIVACY
Managers & Leadership	Weekly 40h Audit & Payroll Accuracy	14 Built-In Verification Checks	100% Local & Private (No Cloud)

1. Executive Summary: What Is This Project?

In our weekly operations, full-time contractors are contracted to fulfill a **40-hour work week**. However, candidate time comes from two completely separate files every single week:

1. **Report 1 (Working Hours):** Timecard records of actual hours worked on the job.
2. **Report 2 (Leaves Details):** HR records of sick days, medical appointments, and approved time off.

Previously, operations staff had to open both spreadsheets in Excel, search for each candidate, read through unstructured doctor notes, convert fractions of days into hours, and calculate the combined sum by hand. This was a recurring manual bottleneck that consumed valuable hours every month and led to costly payroll disputes.

The **Under 40 Report Dashboard** completely automates this process. The user simply **drags and drops both reports** into the browser. In less than **2 seconds**, the tool merges the data, intelligently calculates leave hours from freehand doctor notes, and flags anyone whose total falls **strictly under 40 hours**.

< 2 Sec	100%	Hours Saved	100% Safe
RECONCILIATION TIME	CALCULATION ACCURACY	IN A MONTH	CONFIDENTIAL & PRIVATE

2. Key Problems Solved for the Business

⏱️ Eliminates Manual Excel Reconciliations

Replaces repetitive VLOOKUP formulas, copy-pasting, and manual math with a single drag-and-drop workflow that runs in seconds.

🩺 Reads Freehand Medical & Leave Notes

Intelligently parses messy HR comments (e.g. "*took 4 hours leave*", "*2.5 hrs doctor*", or full days) without human data-entry mistakes.

⚖️ Fast Operational Accountability

Gives managers instant dropdowns to record who is at fault (Client vs. Candidate), assign pay methods, and track informed clients.

🔒 Protects Confidential Health Data

Runs 100% locally on the user's computer. Employee medical records and sick certificates never leave the device or upload to external clouds.

3. Tab-by-Tab Architecture & Role Overview

The dashboard organizes weekly contractor compliance into **5 distinct tabs**, each designed for a specific business purpose and stakeholder. Below is the executive reference matrix:

Tab Name & Icon	Primary Question It Answers	Key Business Process Handled	Primary Stakeholders
📊 **Tab 1: Summary Report**	*"Who fell short of 40 hours, who is at fault, and how will they be paid?"*	Management accountability sheet: log fault (Client/Candidate), select form of pay, add comments, exclude part-timers.	Managers, Dispatch Leads, Payroll Officers
📋 **Tab 2: Under 40 Report**	*"What is the exact mathematical breakdown of each shortfall?"*	Mathematical audit: Original Worked Hours + Leaves Hours + Holidays = Adjusted Total vs 40-hour requirement.	Operations Auditors, Compliance Officers
🗂️ **Tab 3: All Processed**	*"What is the compliance health of the entire company this week?"*	Master roster: Displays full workforce side-by-side with color-coded badges (**Compliant** vs **Under 40h**).	Senior Management, HR Directors
⏱️ **Tab 4: Working Hours**	*"Where did the worked hours numbers come from?"*	Raw timesheet transparency: Instant search and audit of the original clock-in CSV without opening Excel.	Timekeepers, Team Leads
🩺 **Tab 5: Leaves Details**	*"Where did the sick leave and appointment numbers come from?"*	Medical audit trail: Review submitted sick certificates, date ranges, and original doctor comments.	HR Coordinators, Sick Leave Auditors

4. High-Level System Workflow (How Data Flows)

Step 1: Ingestion & Sanitization

The dashboard reads both CSVs, normalizes time formats (e.g., `24:00` → `24:00:00`), and removes test accounts and zero-hour rows.

Step 2: Note Extraction & Summation

Smart text heuristics scan leave comments for partial hours (defaulting to 8h for full days), merging hours by candidate name.

Step 3: Under-40 Compliance Triage

Candidates with ≥ 40 hours are marked Compliant. Candidates with < 40 hours are routed to the Summary and Under 40 action tabs.

Step 4: Decision Logging & Export

Managers enter operational notes, client fault, and payment decisions, then export clean CSVs/PDFs or save the report state.

5. System Validation Engine: 14 Built-In Checks

To guarantee 100% calculation reliability and prevent payroll disputes, the application implements **14 automated validation checks** organized across **4 protective layers**:

📁 Layer 1: File & Header Validations

4 Checks

1. **File Extension Validation:** Strictly verifies that uploaded documents are valid `.csv` files (or `.json` / `.u40` for saved audit files).
2. **Non-Empty File Guard:** Rejects blank or empty uploads before execution, displaying an immediate alert to prevent silent calculation failures.
3. **Report 1 Header Schema Check:** Verifies that the timesheet file contains the mandatory columns ("*Candidate Name*" and "*Working Hours*").
4. **Report 2 Header Schema Check:** Verifies that the leaves file contains the mandatory "*Candidate Name*" column before linking.

👥 Layer 2: Candidate & Eligibility Validations

5 Checks

5. **Blank Name Filtering:** Automatically drops and skips rows where the candidate name is empty or whitespace.
6. **Test & Blacklist Filter:** Automatically identifies and discards test or dummy placeholders (e.g. "*TEST Peter*", "*PENDING SHIFT*").
7. **Role & Title Eligibility Check:** Enforces that candidates hold the required application title: "*ni - employed and fully registered with jobs plus*", filtering out non-billable roles.
8. **Zero-Hours Row Exclusion:** Automatically ignores candidate rows logged with 0 hours or empty clock-in entries.
9. **Part-Timer Exclusion Filter:** Cross-checks candidates against the deleted part-timers list so contracted part-time workers are not audited as shortfalls.

⚙️ Layer 3: Data Parsing & Smart Matching Validations

3 Checks

10. **Time Format Normalization:** Validates messy time inputs (`HH:MM:SS`, `HH:MM`, or decimal hours) and converts them cleanly to standard decimal hours.
11. **Smart Name Matching:** Reconciles candidate identities across both reports even when middle names are expanded or minor spelling variations occur (e.g. "*Martines*" vs "*Martinez*").
12. **Medical Comment & Date Validation:** Scans freehand doctor notes with regex ("*half hour*", "*X mins*", "*X hrs*") and validates `DD/MM/YYYY` date ranges before defaulting to standard 8 hours.

🎯 Layer 4: Business Rules & Compliance Validations

2 Checks

13. **40-Hour Compliance Threshold:** Evaluates combined hours (*Worked + Leaves + Holidays*) against the 40.00-hour requirement to tag compliance.
14. **Manual Override & Input Validation:** Validates user-typed hours and dropdown selections before recalculating weekly balances in real time.

6. Tab 1 Walkthrough: 📊 Summary Report

The **Summary Report** is the primary operational dashboard where managers take action on non-compliant contractors. It filters the entire workforce down strictly to candidates who fell under 40 hours.

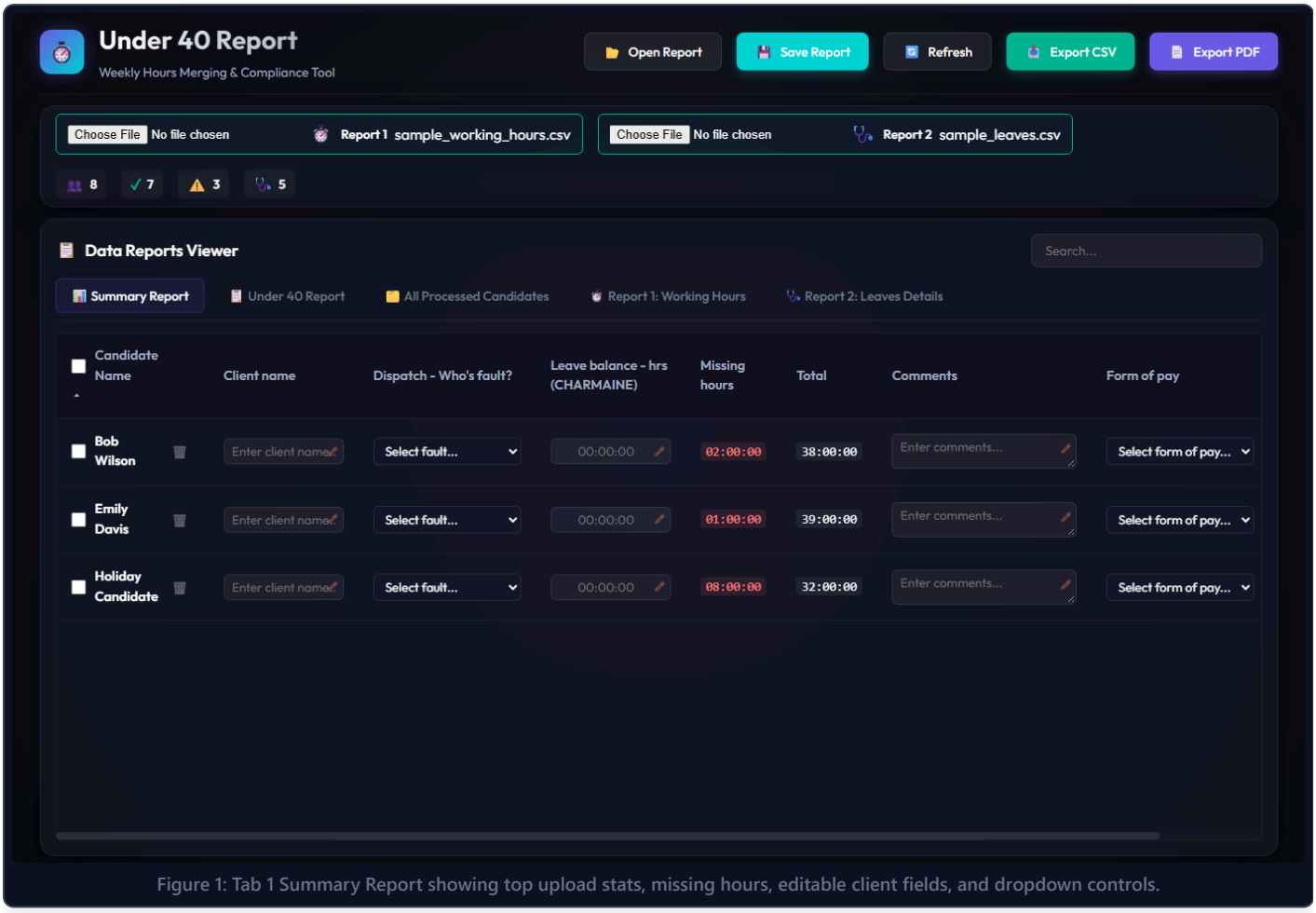


Figure 1: Tab 1 Summary Report showing top upload stats, missing hours, editable client fields, and dropdown controls.

Key Business Processes in Tab 1:

1. Missing Hours & Adjusted Total

Instantly shows how many hours the candidate is short (e.g., *Missing: 02:00:00*, *Total: 38:00:00*) without requiring any manual math.

2. Dispatch - Who's Fault? Dropdown

Allows managers to attribute accountability: **Client** (cancelled shift), **Candidate** (unexcused absence), or **Client+Candidate**.

3. Form of Pay Dropdown

Records payroll instructions: **Paid Leave**, **Unpaid Leave**, **Out of Pocket**, or mixed pay options directly on each row.

4. Leave Balance & Client Notification

Editable inputs allow tracking remaining leave balances in hours and logging whether the client has been notified (**Yes / No**).

Managing Part-Time Contractors & Blank Comments

Fresh Operational Comments: The Comments box in the Summary report starts completely empty (it does not import doctor notes from Report 2). This gives dispatch and managers a clean, dedicated space to document fresh operational explanations regarding why hours were missed.

Part-Time Exclusions: Not every worker under 40 hours is non-compliant; some are contracted as part-timers (e.g. 20h/week). Managers can check the boxes next to part-time workers and click **" 🗑️ Delete Part-Timer(s)"**. This cleanly removes them from the shortfall audit, while keeping them safely stored in a **" ↩️ Part-Timers"** restore drawer in case they need to be re-added later.

7. Detailed Candidate Breakdown & Calculation Visualizer

Clicking on any candidate row in Tab 1 or Tab 2 expands an interactive **Calculation Drawer**. This eliminates confusion by presenting an exact, itemized receipt of how the candidate's weekly hours were calculated.

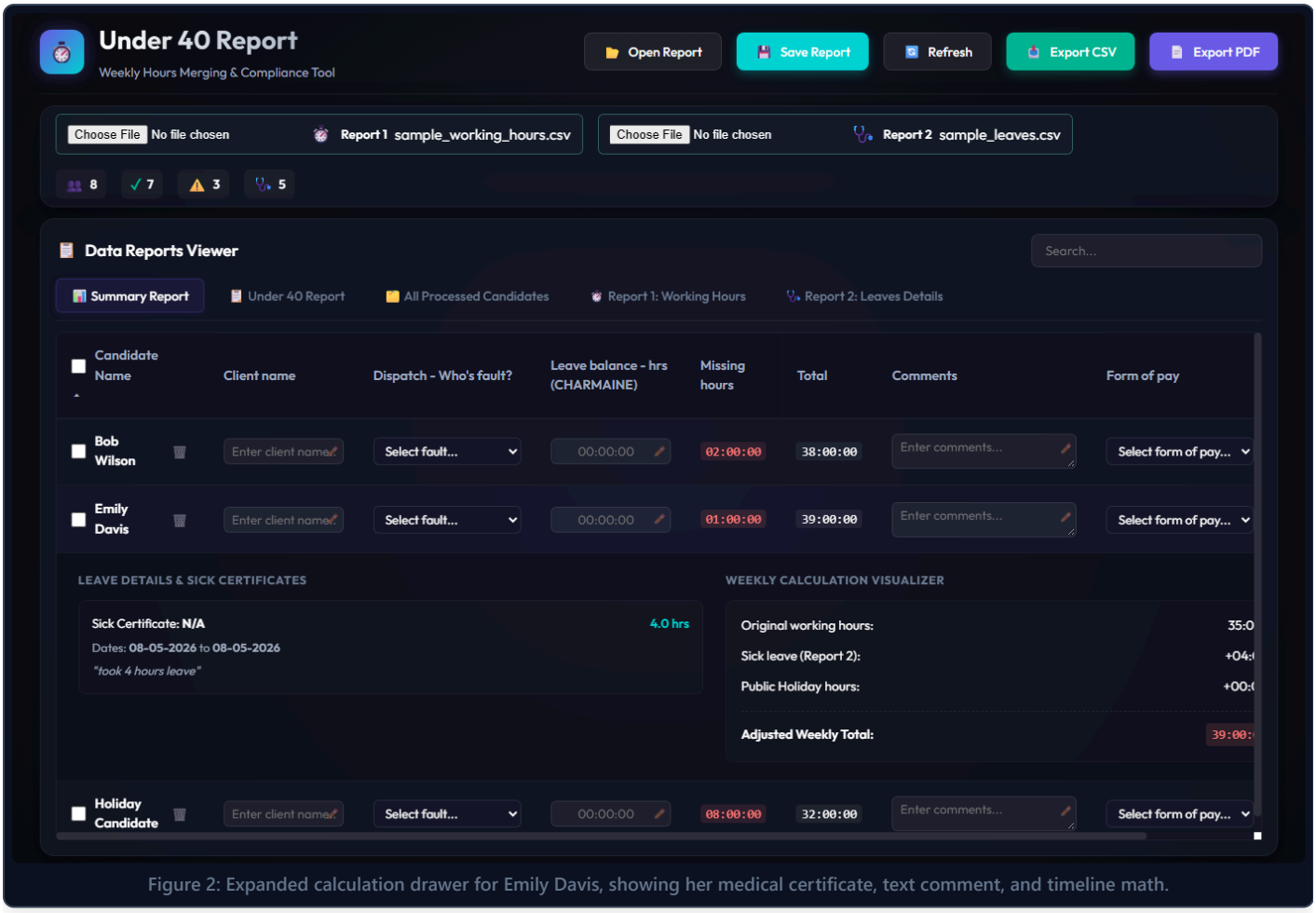


Figure 2: Expanded calculation drawer for Emily Davis, showing her medical certificate, text comment, and timeline math.

Real-World Case Study: Understanding Emily Davis

In messy real-world spreadsheets, candidate **Emily Davis** logged 35 hours of work and submitted a doctor slip with a comment: "*took 4 hours leave*". Here is how the tool reconciled her week:

1. Logged Working Hours (Timecard Report 1):	35:00:00 (35.00 hrs)
2. Approved Leave Hours Extracted from Note ("took 4 hours leave"):	+04:00:00 (4.00 hrs)
3. Public Holiday Credit (Editable override):	+00:00:00 (0.00 hrs)
4. Adjusted Weekly Total (Actual + Leave):	39:00:00 (39.00 hrs)
5. Final Audit Flag: Short of 40h standard requirement by:	**01:00:00 (Missing 1 Hour)**

> **Why This Protects the Business:** In manual Excel workflows, someone skimming the report might miss the freehand note "*took 4 hours leave*" and dock Emily 5 hours of pay. The tool read the note, credited her 4 hours, and correctly identified that she is only 1 hour short—preventing contractor payroll disputes before they happen.

Interactive Leave Overrides

If HR approves an exception, managers can click **"Override Hours"** directly inside the drawer or adjust **Public Holiday** credit. The adjusted total and missing hours recalculate across the entire dashboard instantly.

8. Tab 2 Walkthrough: 📋 Under 40 Report

While Tab 1 is built for manager actions, **Tab 2 (Under 40 Report)** is built for operational auditors. It presents the raw mathematical equation for every candidate who failed to reach 40 hours.

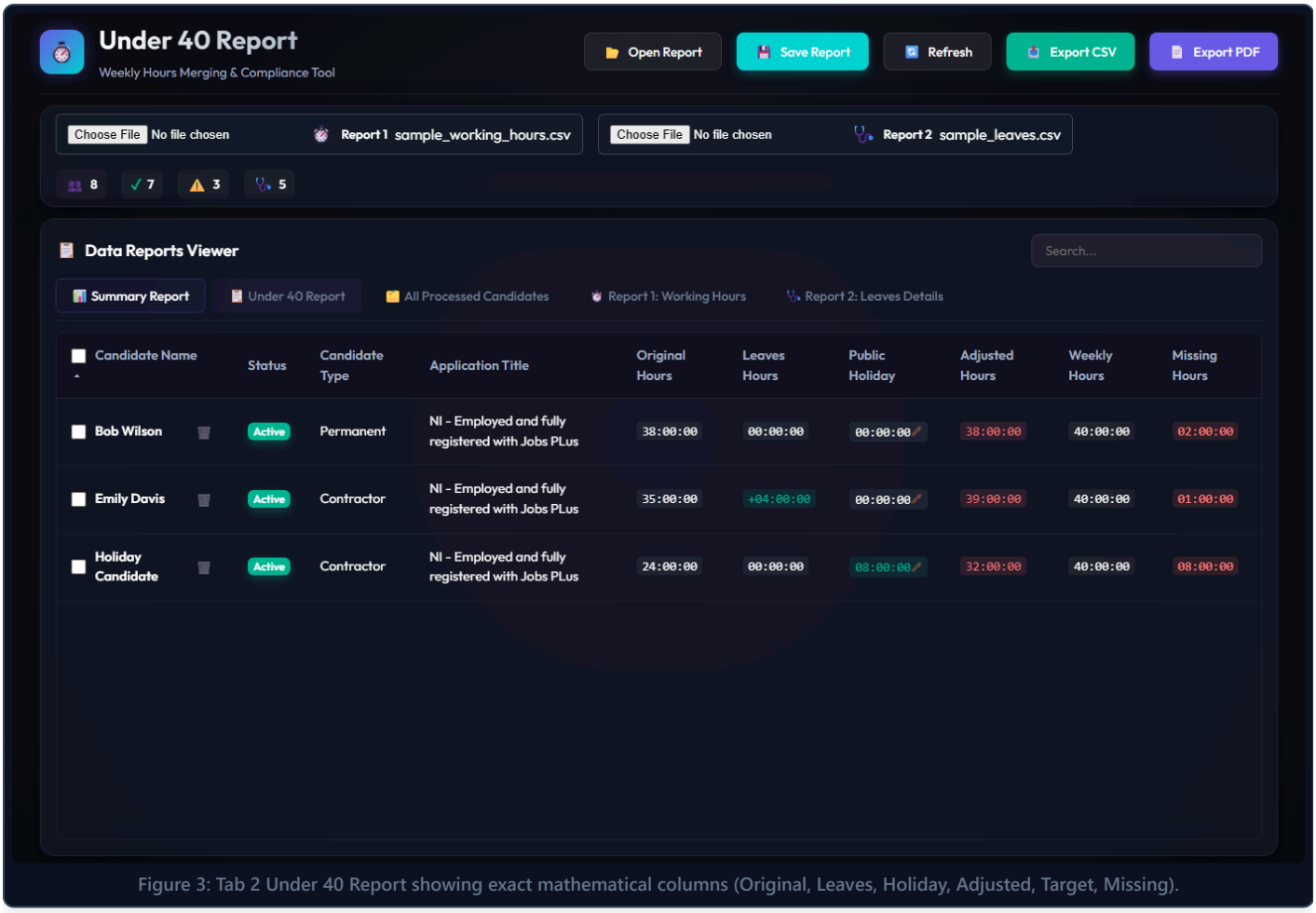


Figure 3: Tab 2 Under 40 Report showing exact mathematical columns (Original, Leaves, Holiday, Adjusted, Target, Missing).

Columns & Data Points Displayed in Tab 2:

- **Candidate Name & Status:** Displays employment status (e.g. *Active*) and job category.
- **Original Hours:** Unadjusted hours worked from timecards (e.g., `35:00:00`).
- **Leaves Hours:** Total approved medical and leave hours added (e.g., `+04:00:00`).
- **Public Holiday:** Editable field to credit statutory company holidays.
- **Adjusted Hours:** Sum of (Worked + Leaves + Holidays).
- **Weekly Hours:** Standard baseline expectation (`40:00:00`).
- **Missing Hours:** The exact compliance deficit highlighted in bold red.

> **Audit Transparency:** Every number is mathematically traceable. If a candidate or client questions an invoice, auditors can reference this exact screen to prove the exact calculation.

9. Tab 3 Walkthrough: 🗂️ All Processed Candidates

Tab 3 (All Processed Candidates) gives leadership the complete 360° view of the entire organization. It displays all active contractors—both those who met compliance and those who fell short.

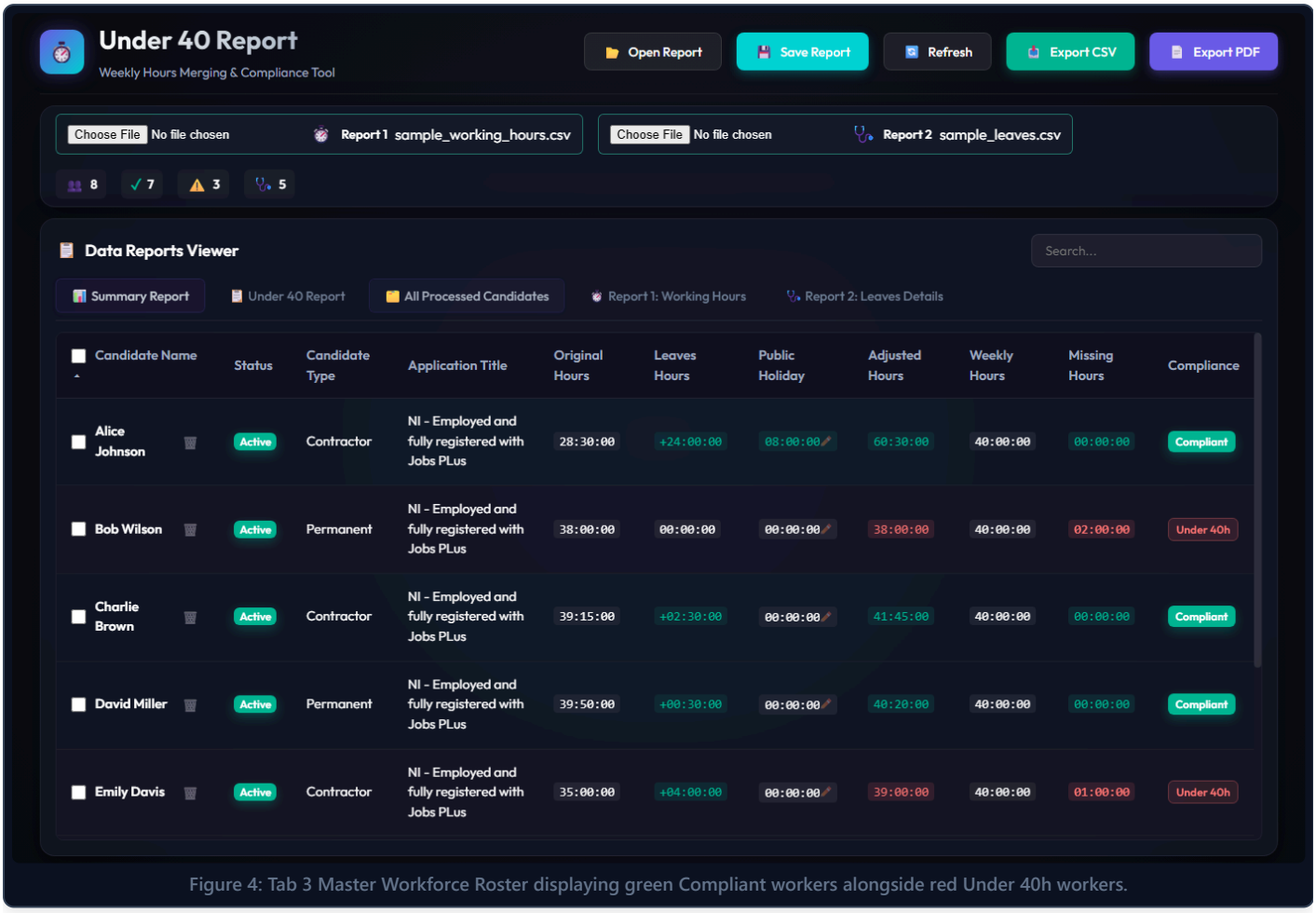


Figure 4: Tab 3 Master Workforce Roster displaying green Compliant workers alongside red Under 40h workers.

Key Features of the Master Roster:

🟢 Compliant Contractors

Candidates who reached 40+ hours are highlighted in green with a **Compliant** badge (e.g. Alice Johnson with 60.5h or Charlie Brown with 41.75h).

🔴 Under 40 Contractors

Candidates who fell short are clearly marked with a red **Under 40h** badge, making workforce exceptions impossible to miss.

🧹 Automatic Data Sanitization

Filters out dummy/test rows (e.g., "*TEST Peter*"), non-billable titles, and zero-hour placeholders, ensuring a pristine executive view.

⚡ Search As You Type

Search any contractor name or role instantly. Even across hundreds of workers, results filter in real-time with zero lag.

10. Tabs 4 & 5 Walkthrough: Raw Source Data Transparency

A core strength of the tool is **100% data transparency**. Tabs 4 and 5 allow users to inspect the unedited source CSVs without ever having to open Excel.

⏱️ Tab 4: Working Hours (Raw Timecards)

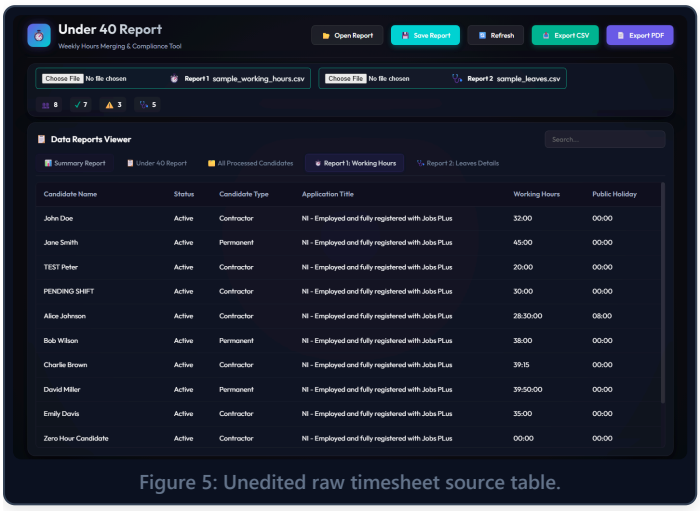


Figure 5: Unedited raw timesheet source table.

Displays the raw hours file exactly as exported from the scheduling system. Allows instant column sorting and row-level verification of shift clock-ins.

🩺 Tab 5: Leaves Details (Raw Medical Slips)

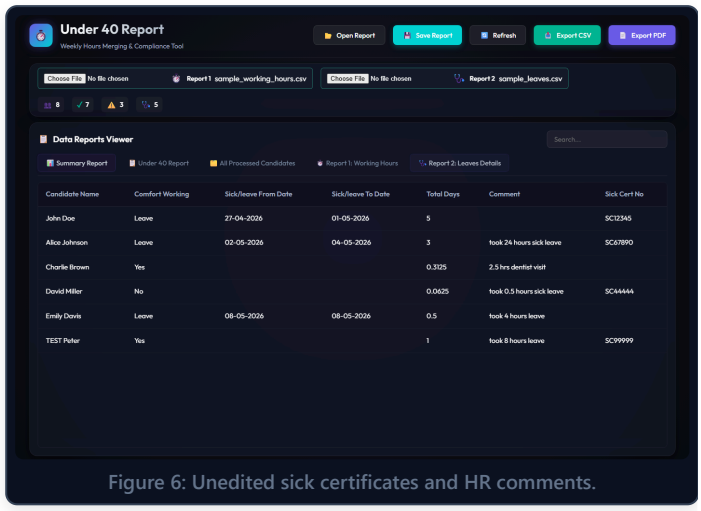


Figure 6: Unedited sick certificates and HR comments.

Displays doctor certificate numbers, leave dates, and unedited comments. Ensures no medical certificates were overlooked during the audit.

11. Header Actions & Business Value Summary

💾 Save & 📂 Open Reports (.json)

Save audit progress on Friday afternoon and reopen it on Monday morning. All manager notes, client faults, and overrides are preserved.

📥 One-Click Export (CSV & PDF)

Generate clean, professional spreadsheets and printable PDF summaries ready to email directly to payroll or executive leadership.

💰 Saves Hours Every Month

Eliminates tedious monthly spreadsheet drudgery, freeing operations staff to focus on high-value client relationships.

🎯 Zero Human Calculation Errors

Guarantees standardized arithmetic and eliminates payroll friction, underpayments, and billing disputes.

> **Conclusion:** The Under 40 Report Dashboard transforms a chaotic, error-prone manual Excel task into a sleek, automated, and audit-proof workflow that saves valuable hours every month and gives management total peace of mind.